FODS LAB PROGRAMS 1-5

1. Scenario: You are working on a project that involves analyzing student performance data for a class of 32 students. The data is stored in a NumPy array named student_scores, where each row represents a student and each column represents a different subject. The subjects are arranged in the following order: Math, Science, English, and History. Your task is to calculate the average score for each subject and identify the subject with the highest average score. Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each student in given order.

ALGORITHM

1. Start
2. Import the NumPy library.
3. Create a 4×4 array named student_scores containing marks for Math, Science, English, and History.
4. Calculate the average of each subject using np.mean(student_scores, axis=0).
5. Store the subject names in a list.
6. Find the index of the highest average using np.argmax().
7. Display the average scores of all subjects.
8. Display the subject with the highest average and its score.
9. Stop

CODE

```
import numpy as np


student_scores = np.array([
    [85, 90, 78, 88],
    [92, 85, 80, 91],
    [78, 88, 95, 82],
    [90, 92, 85, 89]
```

```
])

subjects = ["Math", "Science", "English", "History"]

avg = np.mean(student_scores, axis=0)

for i in range(4):
    print(subjects[i], ":", avg[i])

print("Highest Average:", subjects[np.argmax(avg)])
```

OUTPUT

Math : 86.25

Science : 88.75

English : 84.5

History : 87.5

Highest Average: Science

2.Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a NumPy array. Question: How would you find the average price of all the products sold in the past month? Assume 3x3 matrix with each row representing the sales for a different product

ALGORITHM

1. Start
2. Import the NumPy library.
3. Create a 3×3 array containing sales data for three products.
4. Calculate the average using np.mean(sales).
5. Display the average price.
6. Stop

CODE

```
import numpy as np


sales = np.array([
    [100, 150, 200],
    [120, 180, 240],
    [90, 160, 210]
])


avg_price = np.mean(sales)


print("Average price:", avg_price)
```

OUTPUT

Average price: 161.11111111111111

3. Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price. Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

Algorithm

1. Start
2. Import the NumPy library.
3. Create the house_data array.
4. Select houses with more than four bedrooms using house_data[:, 0] > 4.
5. Extract the sale prices using filtered[:, 2].
6. Calculate the average using np.mean().
7. Display the average sale price.
8. Stop

```
import numpy as np

    # Columns: Bedrooms, Square Footage, Sale Price
    house_data = np.array([
        [3, 1200, 200000],
        [5, 1800, 350000],
        [6, 2200, 450000],
        [4, 1500, 250000],
        [5, 2000, 400000]
    ])

    # Filter houses with more than 4 bedrooms
    filtered = house_data[house_data[:, 0] > 4]

    # Calculate average sale price (3rd column)
    avg_price = np.mean(filtered[:, 2])

    print("Average sale price:", avg_price)
```

OUTPUT

Average sale price: 400000.0

4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named sales_data, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter.

Algorithm

1. Start
2. Import the NumPy library.
3. Create an array containing sales for four quarters.
4. Calculate total annual sales using np.sum().
5. Calculate percentage increase using: [\frac{Q4-Q1}{Q1}\times100]
6. Display total sales and percentage increase.
7. Stop

```
import numpy as np

    sales_data = np.array([10000, 12000, 15000, 18000])

    # Calculate total sales
    total = np.sum(sales_data)
```

```
14.
15.      # Calculate percentage increase
16.      increase = ((sales_data[3] - sales_data[0]) /
17.                  sales_data[0]) * 100
18.
19.      print("Total annual sales:", total)
20.      print("Percentage increase:", increase, "%")
```

OUTPUT

Total annual sales: 55000

Percentage increase: 80.0 %

5. Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named fuel_efficiency, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models. Question: How would you use NumPy arrays and arithmetic operations to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models?

Algorithm

1. Start
2. Import the NumPy library.
3. Create an array containing the fuel efficiency of car models.
4. Calculate the average using np.mean().
5. Calculate percentage improvement between two models using: [\frac{\text{New}-\text{Old}}{\text{Old}}\times100]
6. Display the average fuel efficiency and percentage improvement.
7. Stop

```
8. import numpy as np
9.
10.      fuel_efficiency = np.array([20, 25, 30, 35, 40])
11.
12.      # Calculate average fuel efficiency
13.      avg = np.mean(fuel_efficiency)
14.
15.      # Percentage improvement between two car models
16.      improvement = ((fuel_efficiency[1] - fuel_efficiency[0]) /
17.                     fuel_efficiency[0]) * 100
```

```
18.
19.      print("Average fuel efficiency:", avg, "MPG")
20.      print("Percentage improvement:", improvement, "%")
```

OUTPUT

Average fuel efficiency: 30.0 MPG

Percentage improvement: 25.0 %